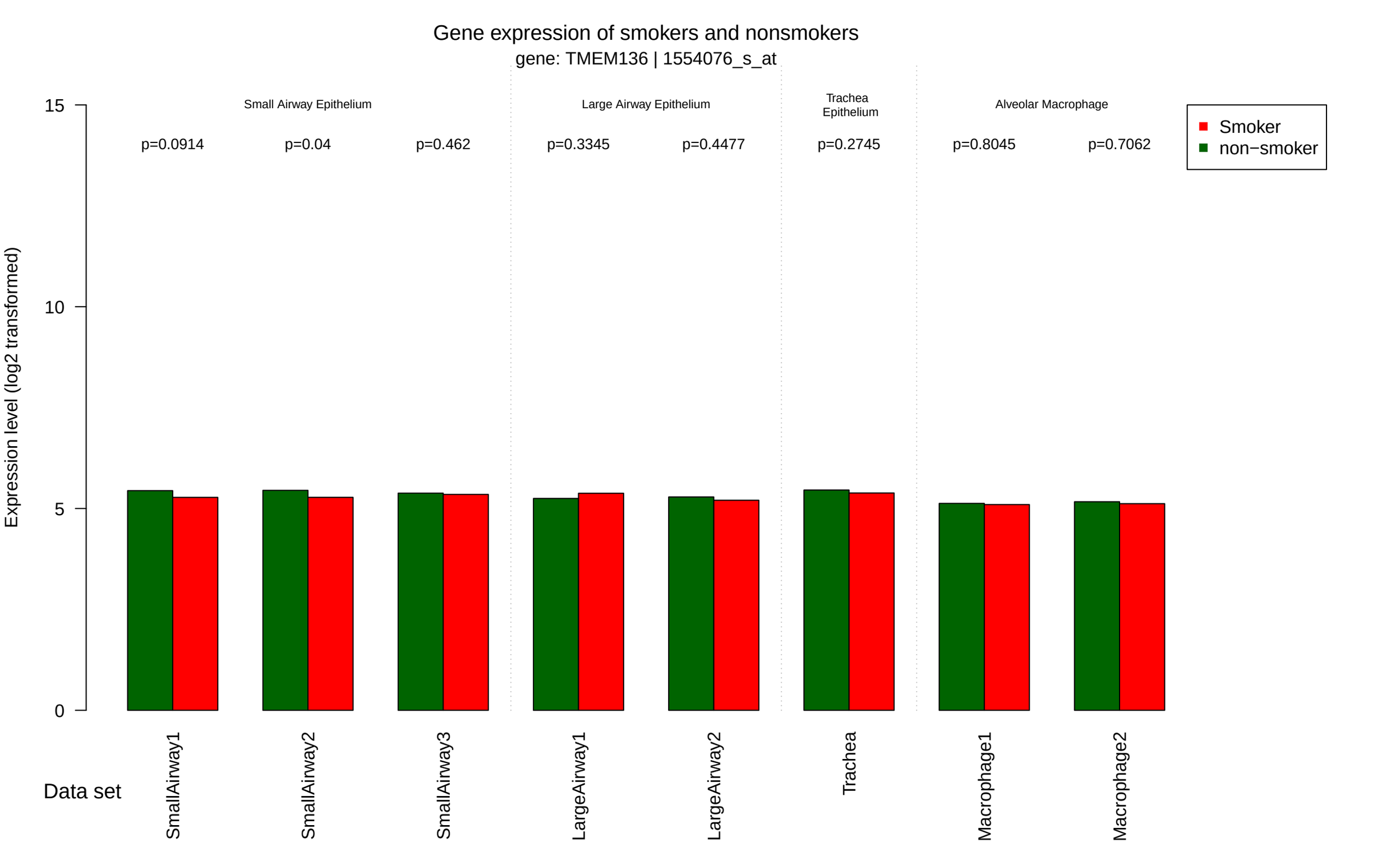
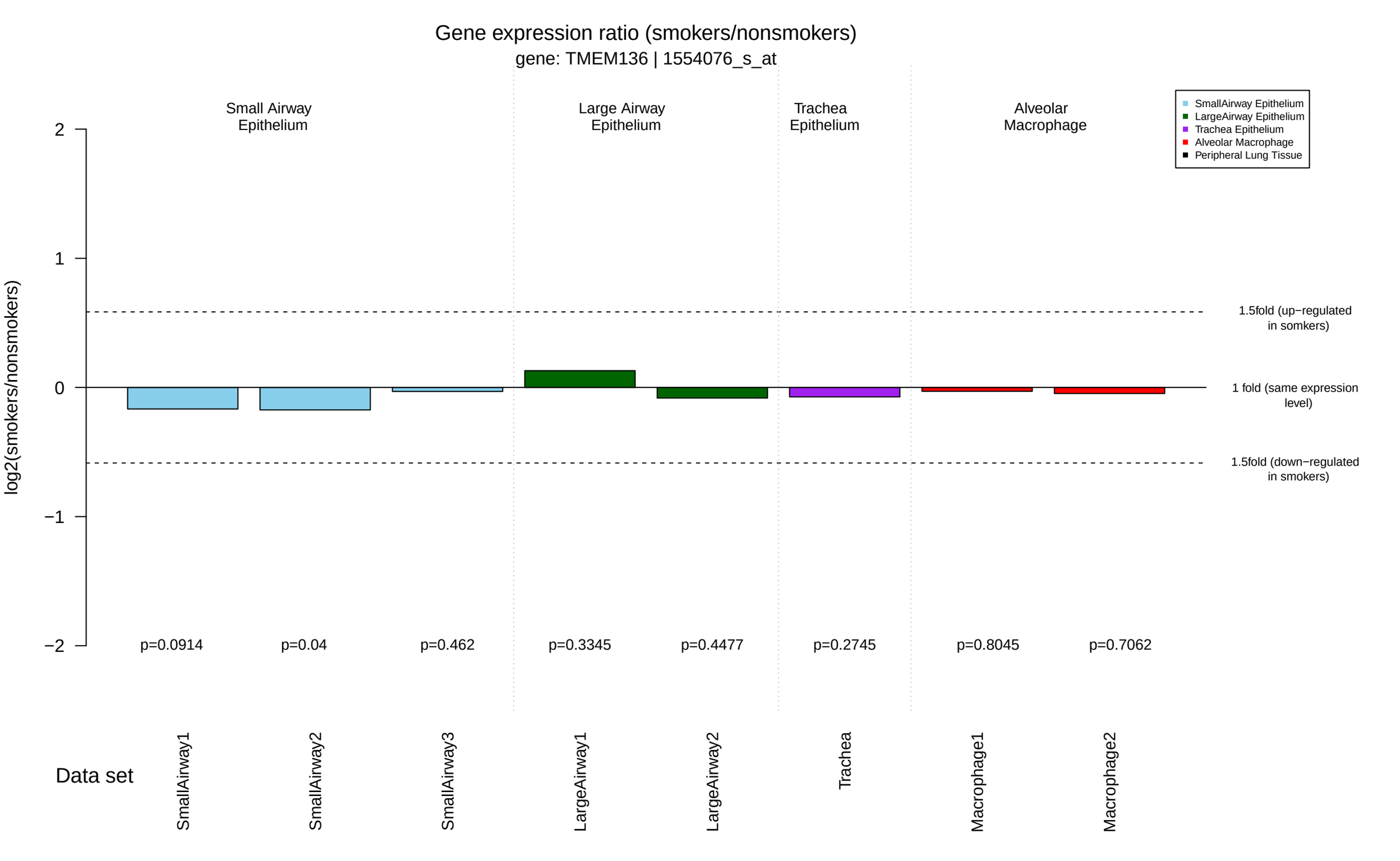
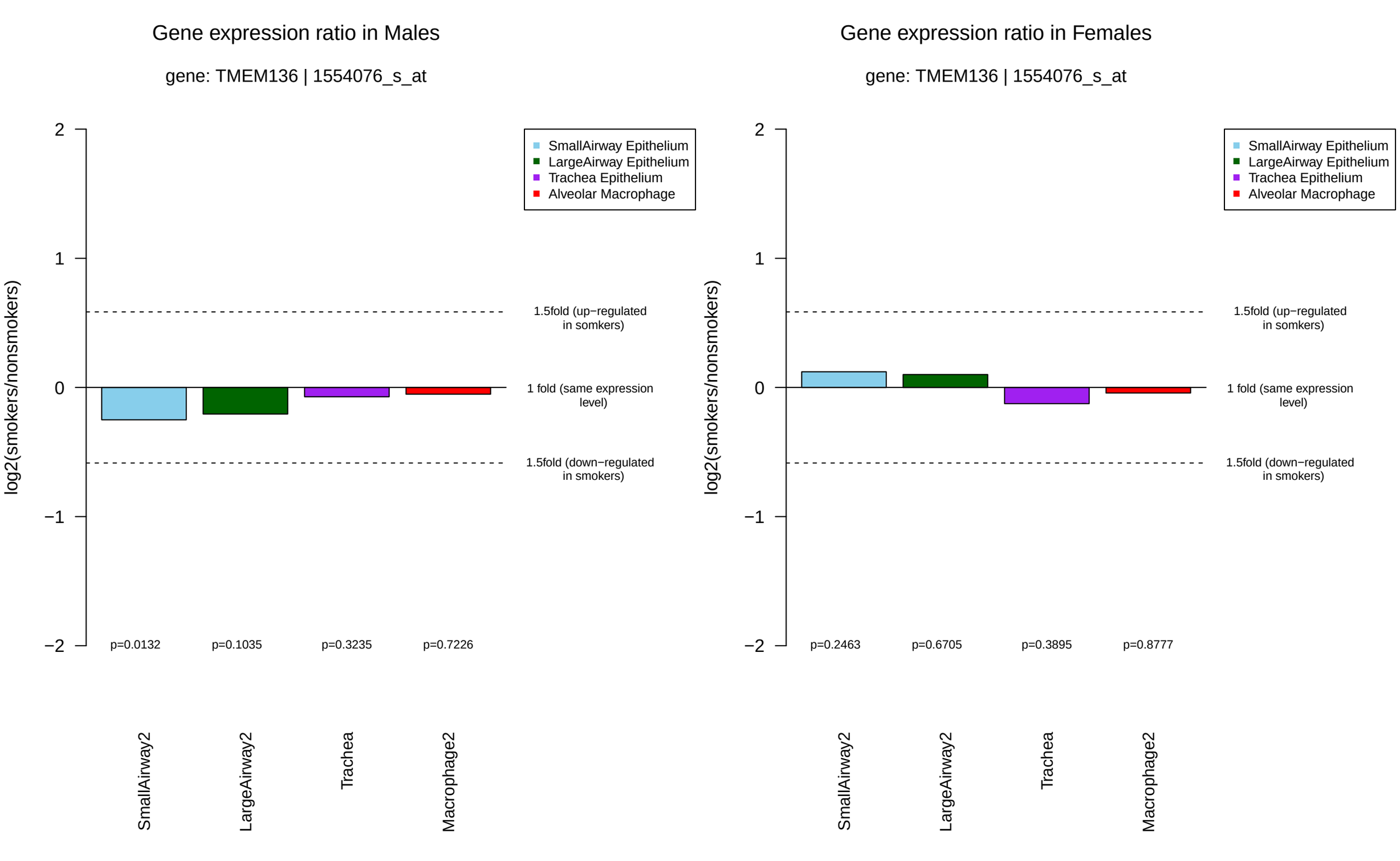
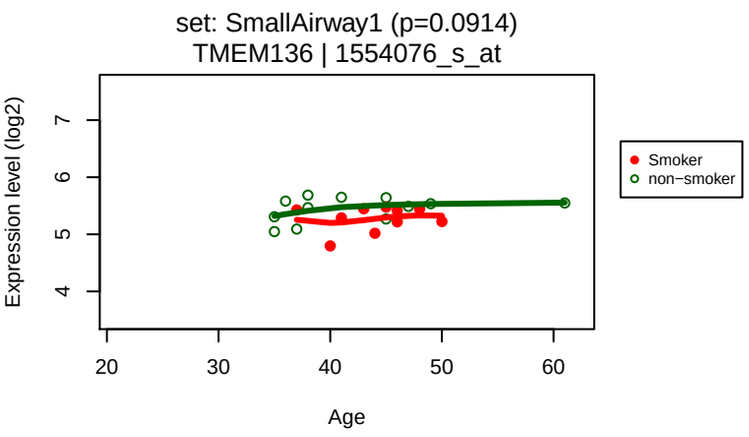
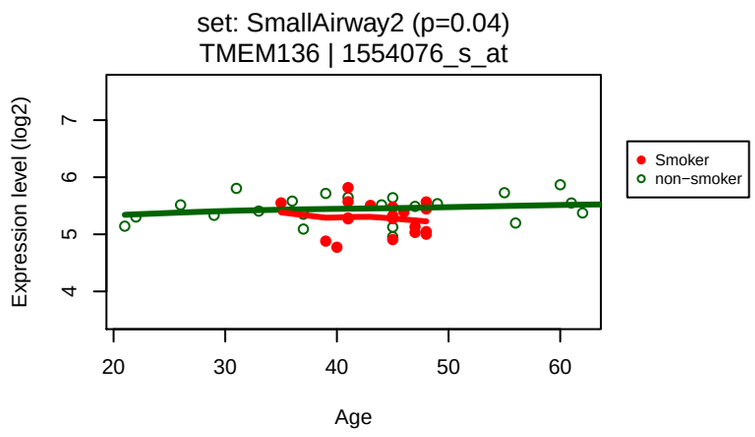
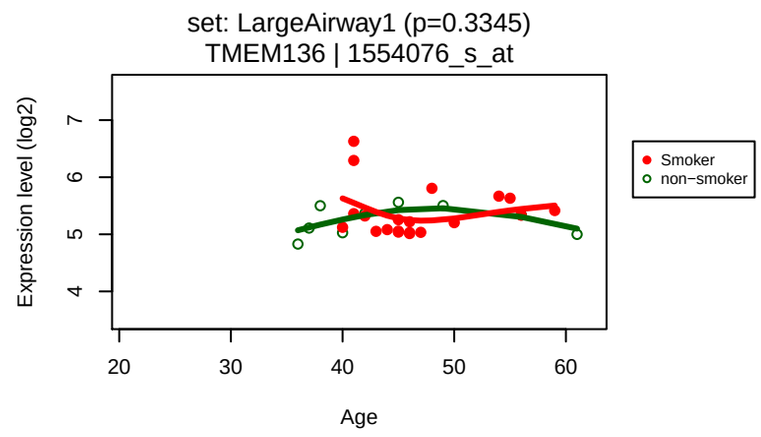

Gene expression of smokers and nonsmokers
gene: TMEM136 | 1554076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0914
p=0.04
p=0.462
p=0.3345
p=0.4477
p=0.2745
p=0.8045
p=0.7062
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

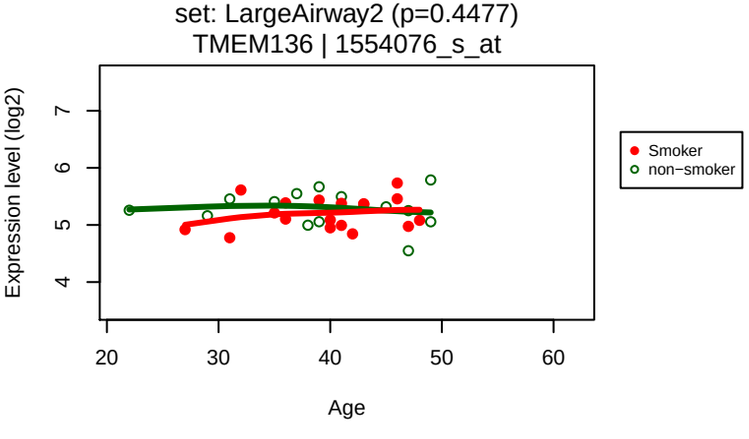

Gene expression ratio (smokers/nonsmokers)
gene: TMEM136 | 1554076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0914
p=0.04
p=0.462
p=0.3345
p=0.4477
p=0.2745
p=0.8045
p=0.7062
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

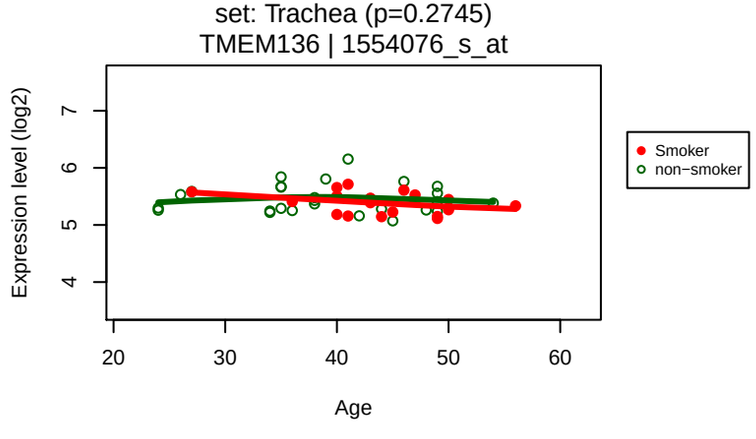

Gene expression ratio in Males
gene: TMEM136 | 1554076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0132
p=0.1035
p=0.3235
p=0.7226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM136 | 1554076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2463
p=0.6705
p=0.3895
p=0.8777
SmallAirway2
LargeAirway2
Trachea
Macrophage2

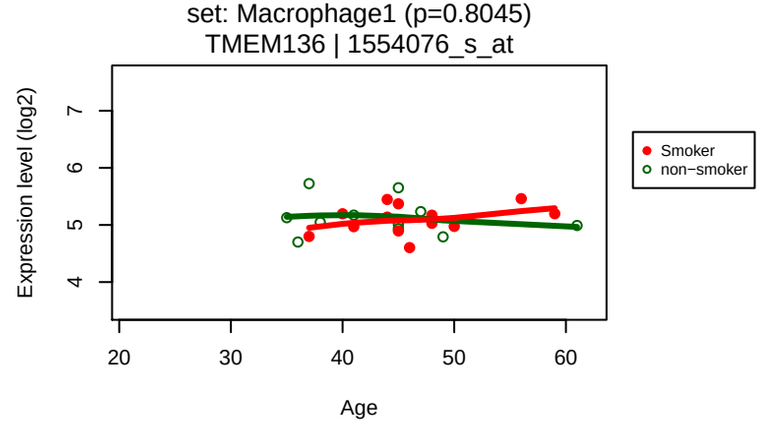

set: SmallAirway1 (p=0.0914)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker

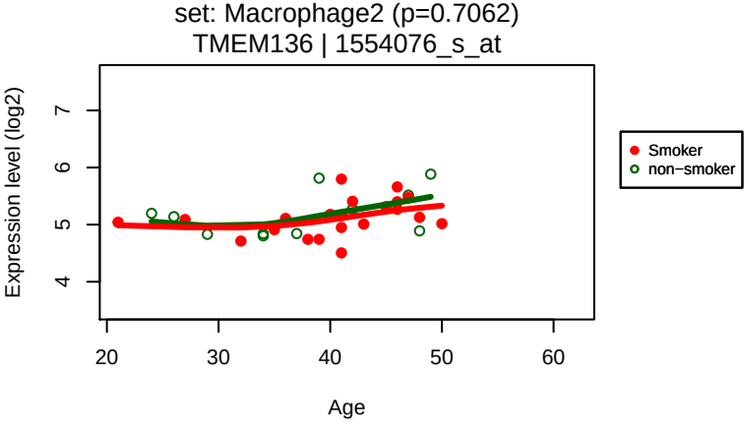

set: SmallAirway2 (p=0.04)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker

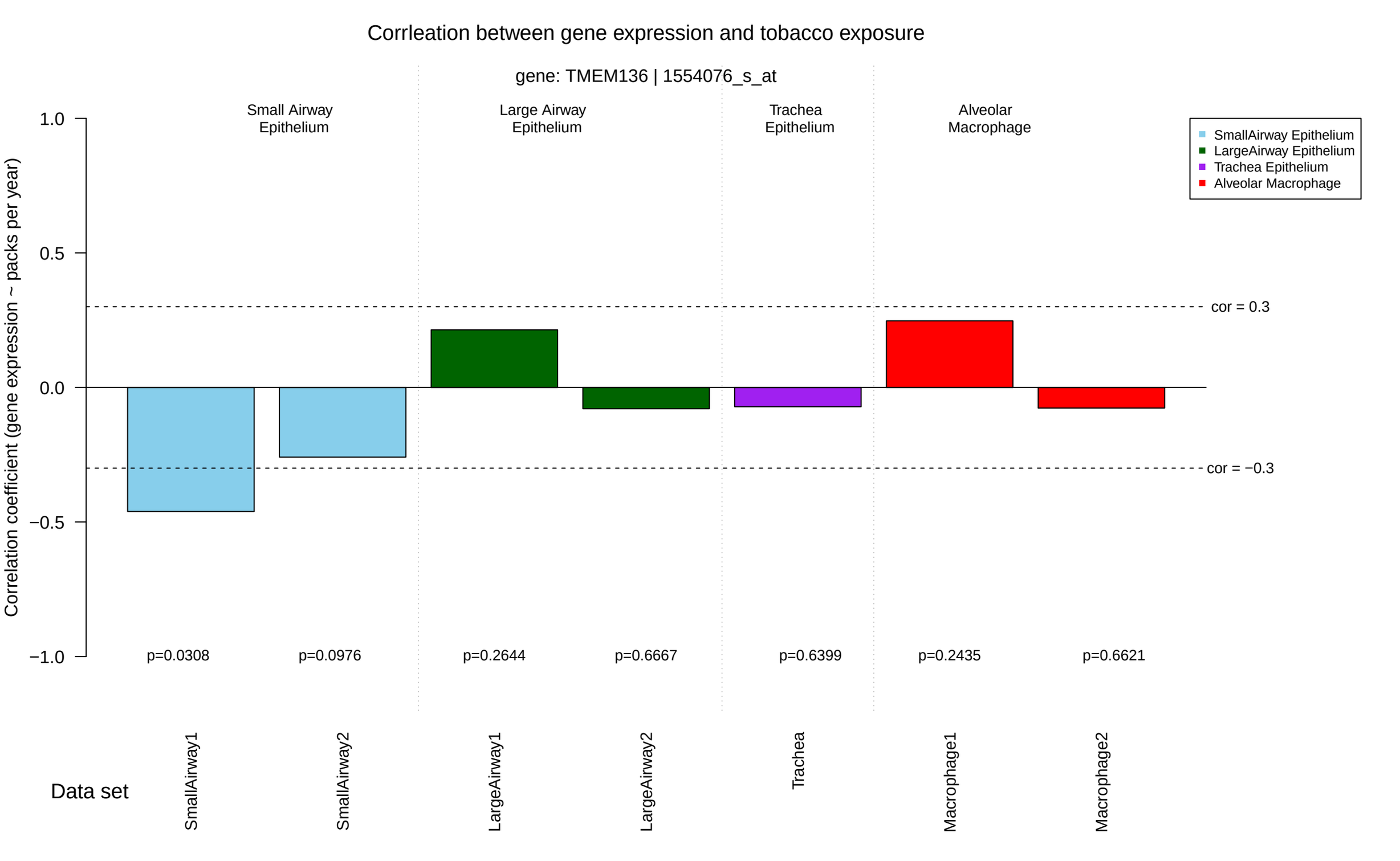

set: LargeAirway1 (p=0.3345)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4477)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2745)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8045)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7062)
TMEM136 | 1554076_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM136 | 1554076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0308
p=0.0976
p=0.2644
p=0.6667
p=0.6399
p=0.2435
p=0.6621
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2